

Table 1. Electrical Performance

Parameter	Symbol	Min.	Typ	Max	Units
Nominal Frequency ¹	F _{NOM}		24.000		MHz
Mode		Fundamental, AT - Cut			
Operating Temperature Range	T _{OP}	-40		125	°C
Total Stability ²	F _{STAB}			±50	ppm
Equivalent Series Resistance	ESR			100	ohms
Load Capacitance	C _L		8		pF
Shunt Capacitance	C _o	-10%	0.83 pF	+10%	
Motional Capacitance	C ₁	-10%	2.88 fF	+10%	
C _o /C ₁		-30%	290	+30%	
Drive Level	DL		10	200	uW
Drive Level Dependence, 50nW-100uW	DLD			2	ohms
Insulation Resistance		500			MOhm
Storage Temperature	T _{STO}	-40		125	°C

Notes:

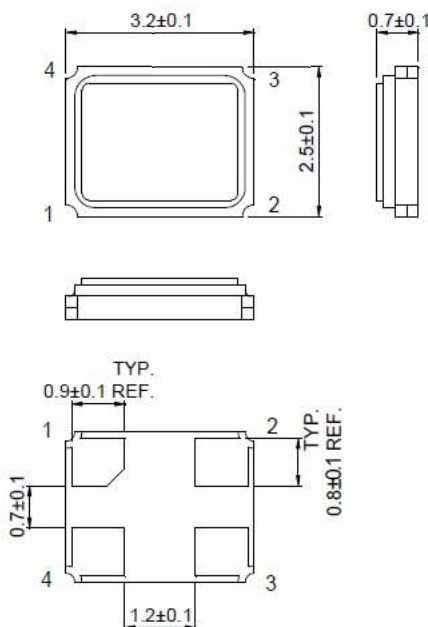
1. Referenced to the Frequency at 25 °C.

2. Total Tolerance includes Initial Accuracy, temperature and 10 years aging at 25°C.

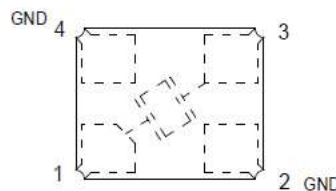
Product is compliant to RoHS directive and fully compatible with lead free assembly.



Package Drawing



All Dimensions in mm



Marking

T240

PDLL

where

T = Manufacturing Location

240 = Frequency, 24.000 MHz

P = Production Control Lot

D = Date Code

LL = Lot Code

Table 2. Pinout

Pin	Function
1	Crystal
2	Connected to cover (Connect to GND)
3	Crystal
4	Connected to cover (Connect to GND)

Table 3. Environmental Compliance

Parameter	Conditions
Drop Test	Height: 120 cm height Number of times: 10 times on a stainless steel plate
Mechanical Shock	Shock Force: Half sine wave, 3000 G Orientation: Three mutually perpendicular axes Number of Times : 3 times each axis Duration: 0.3 msec
Mechanical Vibration	Frequency Range 10-2000 Hz Amplitude: 15.2 mm/10G Sweep Time: 20 minutes Sweep Duration: 4 hours each axes
Solderability	Temperature: 245°C ±5°C Immersion Depth: 1.25 mm Immersion Time: 5 ± 1 seconds Flux: Rosin resin methyl alcohol
Terminal Strength	Mounted on a PCB, shear strength 1.8 kg for 60 seconds
Board Flex	Duration: 60 seconds Deviation: 3 mm
Resistance to Solder Heat	Pre-heat Temp: 125°C Pre heat Time: 60 - 120 seconds Temperature: 260 ± 5°C Duration: 10 ±1 second
High Temperature Storage	Temperature: 125°C ±3°C Duration: 1000 hours
Low Temperature Storage	Temperature: -55°C ±3°C Duration: 1000 hours
Thermal Shock	Temperature: -55/125°C Number of Cycles: 1000 cycles Duration: 30 min soak time, 20 seconds ramp time
Temperature Cycle	Temperature: -55/125°C Number of Cycles: 1000 cycles Duration: 30 minute dwell time
Biased Humidity	Temperature: 85°C ±3°C Humidity: 85% Relative Humidity Duration: 1000 hours
Moisture Sensitivity	MSL1
Contact Pads	Gold over Nickel
Weight	18 mg

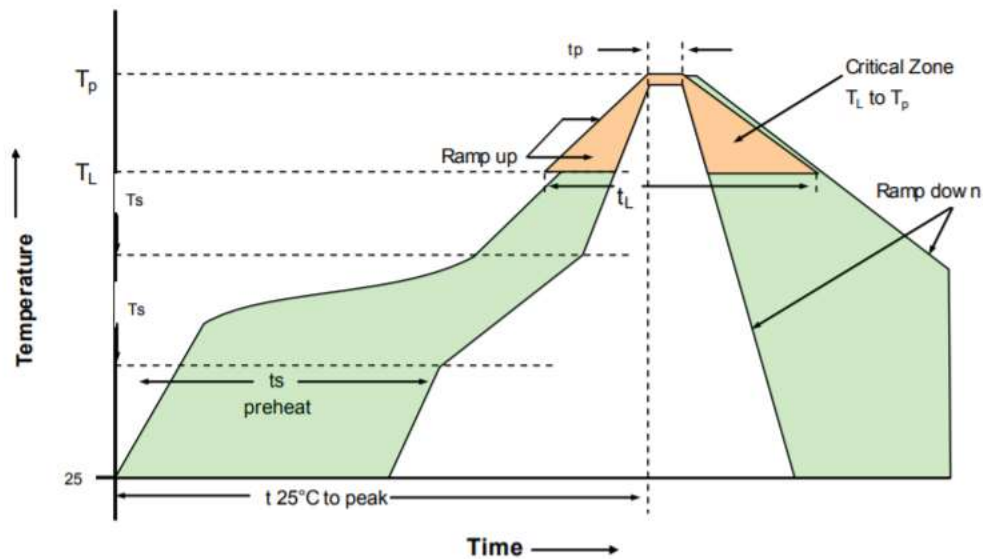


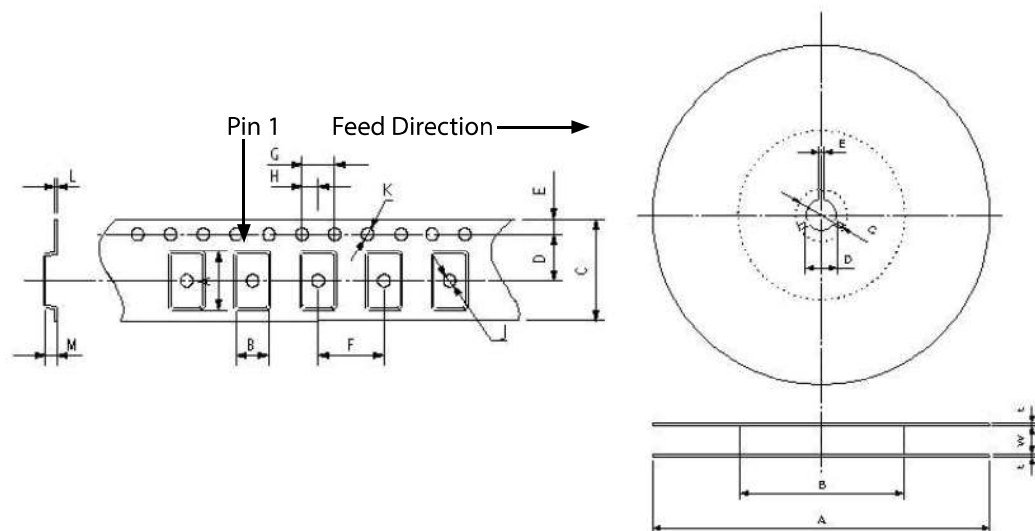
Table 4: Reflow Profile

Parameter	Symbol	Value
PreHeat Time Ts-min Ts-max	t_s	60 sec Min, 260 sec Max 150°C 200°C
Ramp Up	R_{UP}	3 °C/sec Max
Time Above 217 °C	t_L	60 sec Min, 150 sec Max
Time To Peak Temperature	T_{AMB-P}	480 sec Max
Time at 260 °C	t_p	30 sec Max
Ramp Down	R_{DN}	6 °C/sec Max

Tape & Reel

Table 5. Tape and Reel Dimensions (mm)

Tape												Reel							
A	B	C	D	E	F	G	H	J	K	L	M	A	B	C	D	E	W	T	
3.6	2.9	8.0	3.5	1.75	4.0	4.0	2.0	0.5	1.55	0.25	1.65	180	60	21.0	13.0	2.0	9.0	2.0	



3K pieces per reel

Ordering Information

VXM7A - 9001 - 24M0000000xx

Product

3.2 x 2.5mm, Crystal

SCD

Packaging

TR: Tape and Reel

blank: Cut Tape / non TR quantities

Frequency in MHz

Example:

VXM7A-9001-24M0000000TR

Tape and Reel

VXM7A-9001-24M0000000

Cut Tape

Revision History

Revision Date	Approved	Description
Jan 04, 2022	FB	Initial

Contact Information

USA:

100 Watts Street
Mt Holly Springs, PA 17065
Tel: 1.717.486.3411
Fax: 1.717.486.5920

Europe:

Landstrasse
74924 Neckarbischofsheim
Germany
Tel: +49 (0) 7268.801.0
Fax: +49 (0) 7268.801.281



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